Scientists Confirm a Third Type of Magnetism

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For nearly 90 years, scientists sorted magnetic materials into two groups. Now they have strong proof of a third. A research team led by the U.S. Department of Energy's SLAC National Accelerator Laboratory has found convincing experimental evidence for **altermagnetism**, a form of magnetism that mixes traits from both older groups. The study was published in the journal Science. The team studied **manganese telluride**, a material known as MnTe, which had long been classified as an **antiferromagnet**. In it, they found direct evidence of **chirality**, a twist in the material's magnetic structure. It is a twist like that of a spiral staircase. That twist is a key sign of **altermagnetism**. Lead author Daniel Jost of SLAC said the results provide strong confirmation that this third phase is real. Magnetism comes from a property of electrons called spin, often described as up or down. In ferromagnets, the spins line up in the same direction, and together they create the magnetic field of an ordinary magnet. **Antiferromagnets**, identified in 1933, work differently. Neighboring spins point in opposite directions, so their effects cancel out and no overall magnetism appears. Altermagnets are different again. Their spins alternate, like **antiferromagnets**, so they show no overall **magnetization**. Yet they can also carry useful spin currents, a trick linked to ferromagnets. In altermagnets, the crystal patterns around oppositely pointing spins are rotated compared with each other, breaking a **symmetry** found in ordinary **antiferromagnets**. That small difference changes how electrons behave. The term altermagnets dates to 2022, and the new study supplies the missing proof. The discovery could also lead to practical uses, including advanced memory systems, quantum devices, and more energy-efficient data centers.

Word Treasure

altermagnetism -- A third kind of magnetism that mixes traits from the two older kinds.

chirality -- A twist in a material's magnetic structure, like a spiral staircase.

manganese telluride -- The material, called MnTe, that the scientists studied in this experiment.

antiferromagnet -- A magnetic material whose neighboring spins point in opposite directions and cancel out.

magnetization -- The overall magnetic strength that a material shows.

symmetry -- A pattern that stays the same when it is flipped or turned.

Background you need

Magnetism comes from a property of electrons called spin, which is often described as pointing up or down.

In ferromagnets, the spins line up in the same direction and together create the magnetic field of an ordinary magnet, while in antiferromagnets neighboring spins point in opposite directions so their effects cancel out and no overall magnetism appears.

Altermagnets are a third type: their spins alternate like antiferromagnets, so they show no overall magnetization, yet they can also carry useful spin currents like ferromagnets.

The difference comes from the crystal patterns around oppositely pointing spins being rotated compared with each other, which changes how electrons behave.

Break it down

LEAD: Scientists have found strong experimental proof of a third type of magnetism, altermagnetism, after nearly 90 years of sorting magnetic materials into two groups.

KEY EVENT: A research team led by the U.S. Department of Energy's SLAC National Accelerator Laboratory found direct evidence of chirality in manganese telluride, and the study was published in the journal Science.

+- WHO: A research team led by SLAC National Accelerator Laboratory, with lead author Daniel Jost.

+- WHAT: Direct evidence of chirality, a twist in the material's magnetic structure like a spiral staircase, which is a key sign of altermagnetism.

+- WHERE: In manganese telluride (MnTe), a material long classified as an antiferromagnet.

EVIDENCE: In altermagnets the crystal patterns around oppositely pointing spins are rotated compared with each other, breaking a symmetry found in ordinary antiferromagnets, and that small difference changes how electrons behave.

BACKGROUND: In ferromagnets the spins line up in the same direction and together create an ordinary magnet's field, while in antiferromagnets, identified in 1933, neighboring spins point in opposite directions so their effects cancel out.

FUTURE USES: The discovery could lead to advanced memory systems, quantum devices, and more energy-efficient data centers.

Why it matters

Altermagnets could make the memory chips and data centers behind everyday devices faster and use less energy, so this discovery may shape the technology that powers the computers and gadgets children use every day.

Test what you remember

1. What did the research team led by SLAC find convincing experimental evidence for?

- (A) A stronger form of ferromagnetism
- (B) A third type of magnetism called altermagnetism
- (C) A way to turn antiferromagnets into ferromagnets
- (D) A new kind of ordinary magnet

2. Which material did the team study to find evidence of altermagnetism?

- (A) Manganese telluride (MnTe)
- (B) Iron
- (C) Cobalt
- (D) Nickel

3. What is chirality, the key sign of altermagnetism found in the material?

- (A) A twist in the material's magnetic structure, like that of a spiral staircase
- (B) A change in the material's color when it is heated
- (C) A crack in the crystal that blocks electric current
- (D) A vibration of atoms that produces heat

4. In what year were antiferromagnets identified?

- (A) 1933
- (B) 2022
- (C) 2015
- (D) 1989

5. How are altermagnets different from ordinary antiferromagnets?

- (A) Their spins all point in the same direction and create a strong magnetic field
- (B) Their spins point in random directions and cancel each other out completely
- (C) They have no electrons and no magnetic properties at all
- (D) Their spins alternate so they show no overall magnetization, yet they can carry useful spin currents

6. What practical uses could the discovery lead to?

- (A) New kinds of batteries for electric cars only
- (B) Faster airplanes and quieter car engines
- (C) Advanced memory systems, quantum devices, and more energy-efficient data centers
- (D) Stronger refrigerator magnets and compass needles

Answer key (try the quiz first!): 1: B 2: A 3: A 4: A 5: D 6: C

Questions to think about

1. Lead author Daniel Jost of SLAC: Said the results provide strong confirmation that this third phase of magnetism is real.

MY THOUGHTS (at least 20 words):